

ORBES CODE - 01

PHYSICAL PRINT TEST SHEETS

SAMPLE - NOT VALID · EVERY CODE HERE IS SIGNED WITH THE PUBLIC SAMPLE KEY · NEVER AUTHENTIC

CONTENTS

PAGE	RENDITION	SUBSTRATE · INK	SAMPLE CODE
2	BLACK ON WHITE	#FFFFFF · #0A0A0A	O26-J-900001
3	WHITE ON BLACK	#0A0A0A · #FFFFFF	O26-J-900002
4	IVORY PAPER	#F6F2EA · #111111	O26-J-900003
5	MATTE PAPER · GREY 92 %	#EBEBEB · #0A0A0A	O26-J-900004
6	TEXTURED PAPER	#F7F4EC · #141414 · PAPER FIBRES	O26-J-900005
7	BLACK LEATHER	#211C19 · #E9E3D7 · LEATHER GRAIN	O26-J-900006
8	METALLIC SURFACE	#ADADAD · #444444 · BRUSHED METAL	O26-J-900007

SIZES 10 · 15 · 20 · 25 · 30 · 40 · 50 MM = SIDE OF THE PRINTED SQUARE, QUIET ZONE INCLUDED (50 U) · DISTANCES 8 · 10 · 12 · 15 · 20 · 25 CM

1 · PRINT

- PRINT AT 100 % (ACTUAL SIZE). DISABLE FIT TO PAGE, SHRINK TO FIT AND ANY SCALING IN THE PRINTER DRIVER.
- MEASURE THE 100 MM RULER BELOW AND THE 10 MM SCALE BAR AT THE FOOT OF EVERY PAGE: THEY MUST MEASURE 100 MM ± 0.5 AND 10 MM ± 0.1. OTHERWISE DISCARD THE PRINT: EVERY SIZE ON IT IS WRONG.
- USE A LASER OR INKJET PRINTER AT 600 DPI OR MORE ON MATTE PAPER, BEST QUALITY, NO TONER SAVING. RECORD PRINTER, PAPER AND DPI IN EACH TABLE.
- PAGES 2 TO 8 SIMULATE SUBSTRATES ON PAPER. THEY DO NOT REPLACE TESTS ON REAL LEATHER OR METAL: PRODUCE THOSE FROM THE SVG FILES OF THIS KIT.

2 · PREPARE

- CUT EACH TAG OUT ALONG ITS CROP MARKS, OR COVER EVERY OTHER CODE WITH PLAIN PAPER: THE SCANNER READS WHATEVER CODE IS INSIDE THE ORBIT, AND AT 25 CM THE ORBIT SEES THE WHOLE PAGE.
- LAY THE TAG FLAT ON A PLAIN SURFACE UNDER DIFFUSE INDOOR LIGHT (300 TO 500 LUX), NO SPOTLIGHT OR REFLECTION ON THE TAG.

3 · SCAN

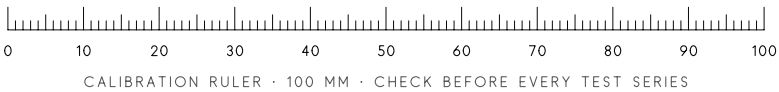
- OPEN /VERIFY ON THE PHONE OVER HTTPS, START SCANNING AND ALLOW THE CAMERA.
- HOLD THE PHONE PARALLEL TO THE TAG AT EACH DISTANCE: 8, 10, 12, 15, 20 AND 25 CM FROM LENS TO TAG. USE A RULER OR A SPACER. KEEP THE CODE CENTRED IN THE ORBIT.
- MAKE 3 ATTEMPTS PER SIZE AND DISTANCE. AN ATTEMPT SUCCEEDS WHEN ANY RESULT SCREEN APPEARS WITHIN 10 S. THESE ARE SAMPLE CODES SIGNED WITH THE PUBLIC SAMPLE KEY: THE RESULT MUST BE INVALID SIGNATURE OR UNKNOWN. IF A SAMPLE CODE IS EVER SHOWN AS AUTHENTIC, STOP AND REPORT IT AS A SECURITY DEFECT.
- REPEAT THE SERIES WITH 2X ZOOM WHEN THE SCANNER SHOWS ITS ZOOM CONTROL (BROWSER DEPENDENT).

4 · RECORD

- WRITE THE NUMBER OF SUCCESSFUL ATTEMPTS IN THE TABLE OF EACH PAGE: ONE TABLE PER PHONE AND PRINT, COPY THE PAGE FOR MORE PHONES. NOTE PHONE MODEL, OS, BROWSER, LIGHT AND PRINTER.

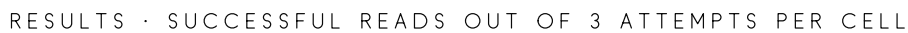
5 · DECIDE

- A SIZE PASSES ON A PHONE WHEN IT READS 3/3 AT TWO NEIGHBOURING DISTANCES WITHOUT ZOOM: REAL USERS DO NOT HOLD AN EXACT DISTANCE. THIS IS THE ROBUST CRITERION OF THE SIMULATION.
- COMPARE WITH DOCS/REPORTS/PRINT-SIZE-MATRIX.MD, A CAMERA SIMULATION. A MINIMUM PRINT SIZE IS VALIDATED ONLY WHEN THE PHYSICAL RESULTS ON REAL PHONES (ANDROID, IPHONE, IPHONE PRO) AGREE WITH IT. WHEN THEY DISAGREE, THE PHYSICAL RESULT WINS.



RENDITION 1 / 7

SAMPLE CODE 026-J-900001 · KEY ID 1 · PUBLIC SAMPLE KEY · A VERIFIER MUST ANSWER INVALID SIGNATURE OR UNKNOWN



PHONE / OS _____ BROWSER _____ DATE _____ LIGHT _____ PRINTER / DPI _____

[illegible]

WRITE 3/3, 2/3, 1/3 OR 0/3 · LEAVE BLANK IF NOT TESTED · WRITE > WHEN THE CODE DOES NOT FIT INSIDE THE ON-SCREEN ORBIT AT THAT DISTANCE
DISTANCE = PHONE LENS TO TAG, PHONE PARALLEL TO THE TAG · A READ = ANY RESULT SCREEN WITHIN 10 S · CUT THE TAG OUT OR MASK THE OTHER CODES FIRST
THIS PAGE SIMULATES WHITE PAPER, CARD AND CERTIFICATES ON PAPER · CONFIRM ON THE REAL MATERIAL BEFORE FIXING A MINIMUM SIZE

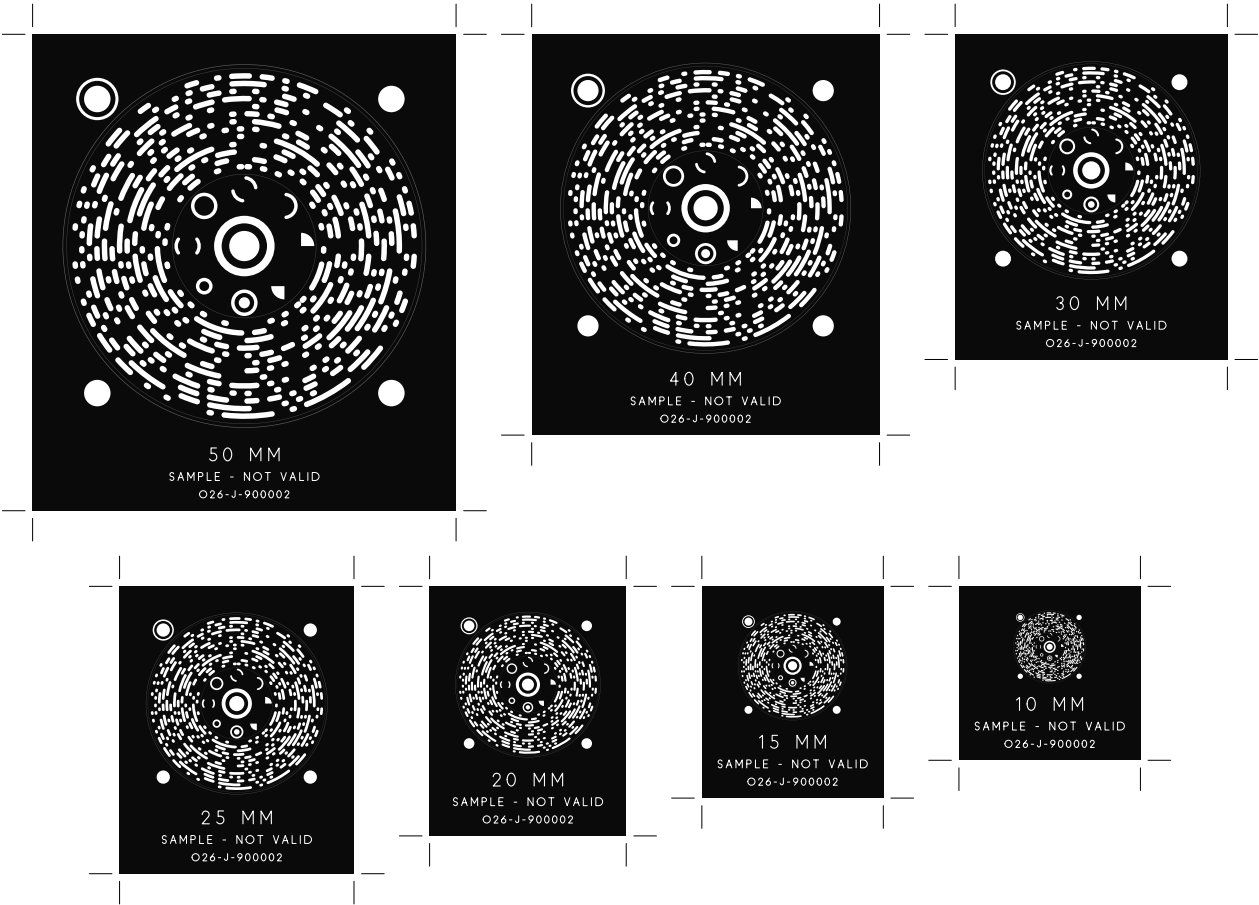


WHITE ON BLACK

RENDITION 2 / 7

INVERTED POLARITY · WHITE ON #0A0A0A

SAMPLE CODE O26-J-900002 · KEY ID 1 · PUBLIC SAMPLE KEY · A VERIFIER MUST ANSWER INVALID SIGNATURE OR UNKNOWN



RESULTS · SUCCESSFUL READS OUT OF 3 ATTEMPTS PER CELL

PHONE / OS _____ BROWSER _____ DATE _____ LIGHT _____ PRINTER / DPI _____

DISTANCE	1X · NO ZOOM						2X ZOOM · IF THE SCANNER OFFERS IT					
	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM
10 MM												
15 MM												
20 MM												
25 MM												
30 MM												
40 MM												
50 MM												

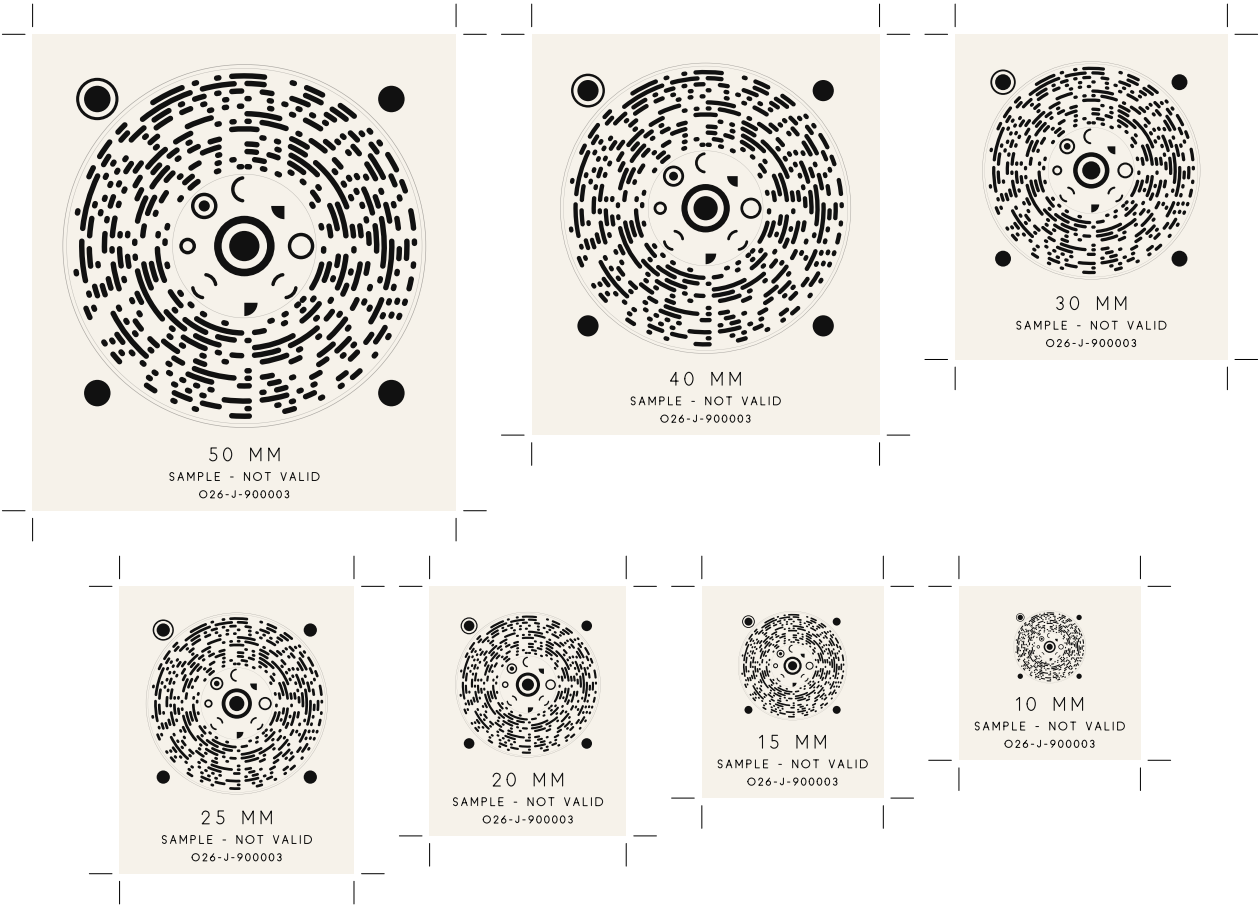
WRITE 3/3, 2/3, 1/3 OR 0/3 · LEAVE BLANK IF NOT TESTED · WRITE > WHEN THE CODE DOES NOT FIT INSIDE THE ON-SCREEN ORBIT AT THAT DISTANCE
DISTANCE = PHONE LENS TO TAG, PHONE PARALLEL TO THE TAG · A READ = ANY RESULT SCREEN WITHIN 10 S · CUT THE TAG OUT OR MASK THE OTHER CODES FIRST
THIS PAGE SIMULATES BLACK CARD WITH WHITE PRINT OR FOIL ON PAPER · CONFIRM ON THE REAL MATERIAL BEFORE FIXING A MINIMUM SIZE

IVORY PAPER

RENDITION 3 / 7

SOFT BLACK #111111 ON IVORY #F6F2EA

SAMPLE CODE O26-J-900003 · KEY ID 1 · PUBLIC SAMPLE KEY · A VERIFIER MUST ANSWER INVALID SIGNATURE OR UNKNOWN



RESULTS · SUCCESSFUL READS OUT OF 3 ATTEMPTS PER CELL

PHONE / OS _____ BROWSER _____ DATE _____ LIGHT _____ PRINTER / DPI _____

DISTANCE	1X · NO ZOOM						2X ZOOM · IF THE SCANNER OFFERS IT					
	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM
10 MM												
15 MM												
20 MM												
25 MM												
30 MM												
40 MM												
50 MM												

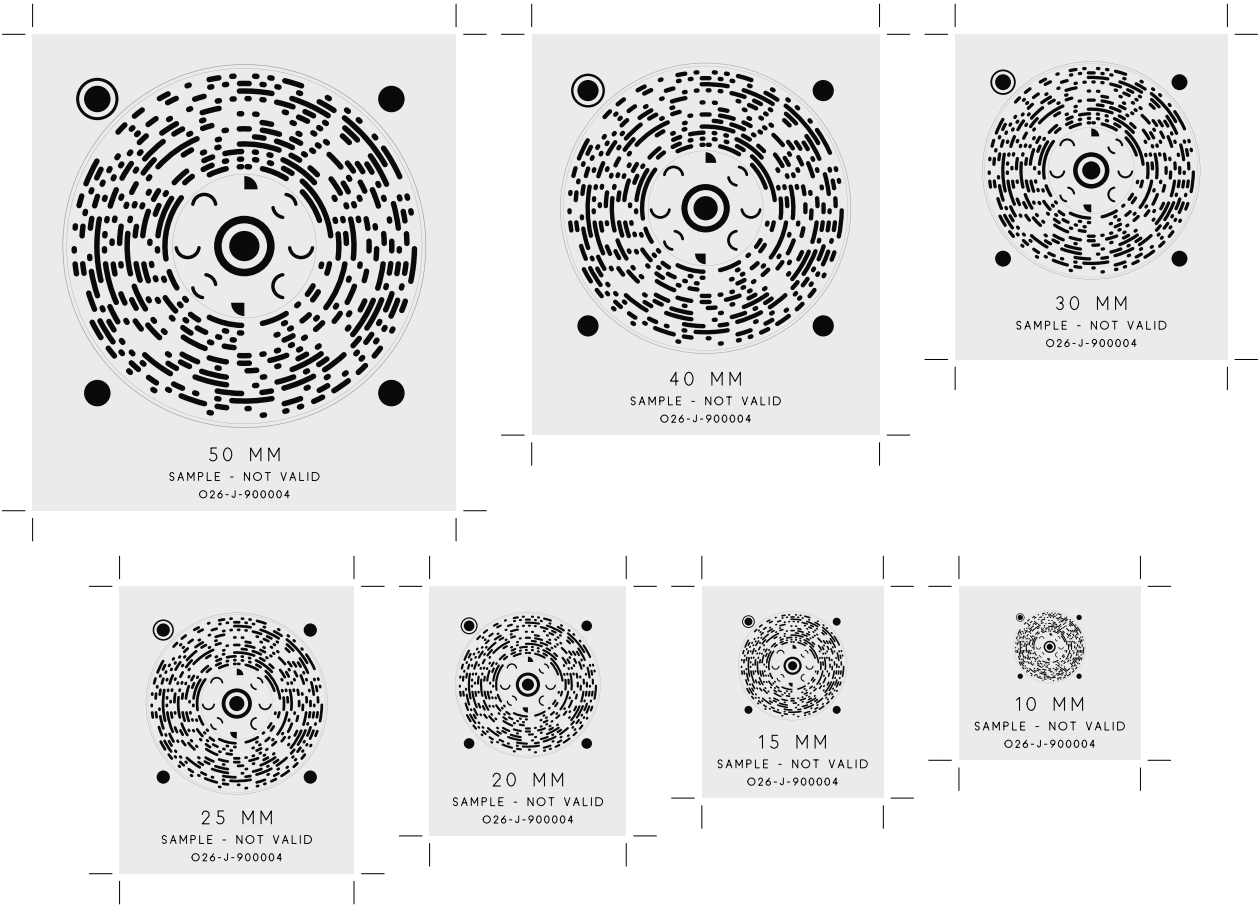
WRITE 3/3, 2/3, 1/3 OR 0/3 · LEAVE BLANK IF NOT TESTED · WRITE > WHEN THE CODE DOES NOT FIT INSIDE THE ON-SCREEN ORBIT AT THAT DISTANCE
DISTANCE = PHONE LENS TO TAG, PHONE PARALLEL TO THE TAG · A READ = ANY RESULT SCREEN WITHIN 10 S · CUT THE TAG OUT OR MASK THE OTHER CODES FIRST
THIS PAGE SIMULATES IVORY CARD STOCK, HANG TAGS, CERTIFICATES ON PAPER · CONFIRM ON THE REAL MATERIAL BEFORE FIXING A MINIMUM SIZE

MATTE PAPER · GREY 92 %

RENDITION 4 / 7

BLACK ON 92 % GREY #EBEBEB

SAMPLE CODE O26-J-900004 · KEY ID 1 · PUBLIC SAMPLE KEY · A VERIFIER MUST ANSWER INVALID SIGNATURE OR UNKNOWN



RESULTS · SUCCESSFUL READS OUT OF 3 ATTEMPTS PER CELL

PHONE / OS _____ BROWSER _____ DATE _____ LIGHT _____ PRINTER / DPI _____

DISTANCE	1X · NO ZOOM						2X ZOOM · IF THE SCANNER OFFERS IT					
	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM
10 MM												
15 MM												
20 MM												
25 MM												
30 MM												
40 MM												
50 MM												

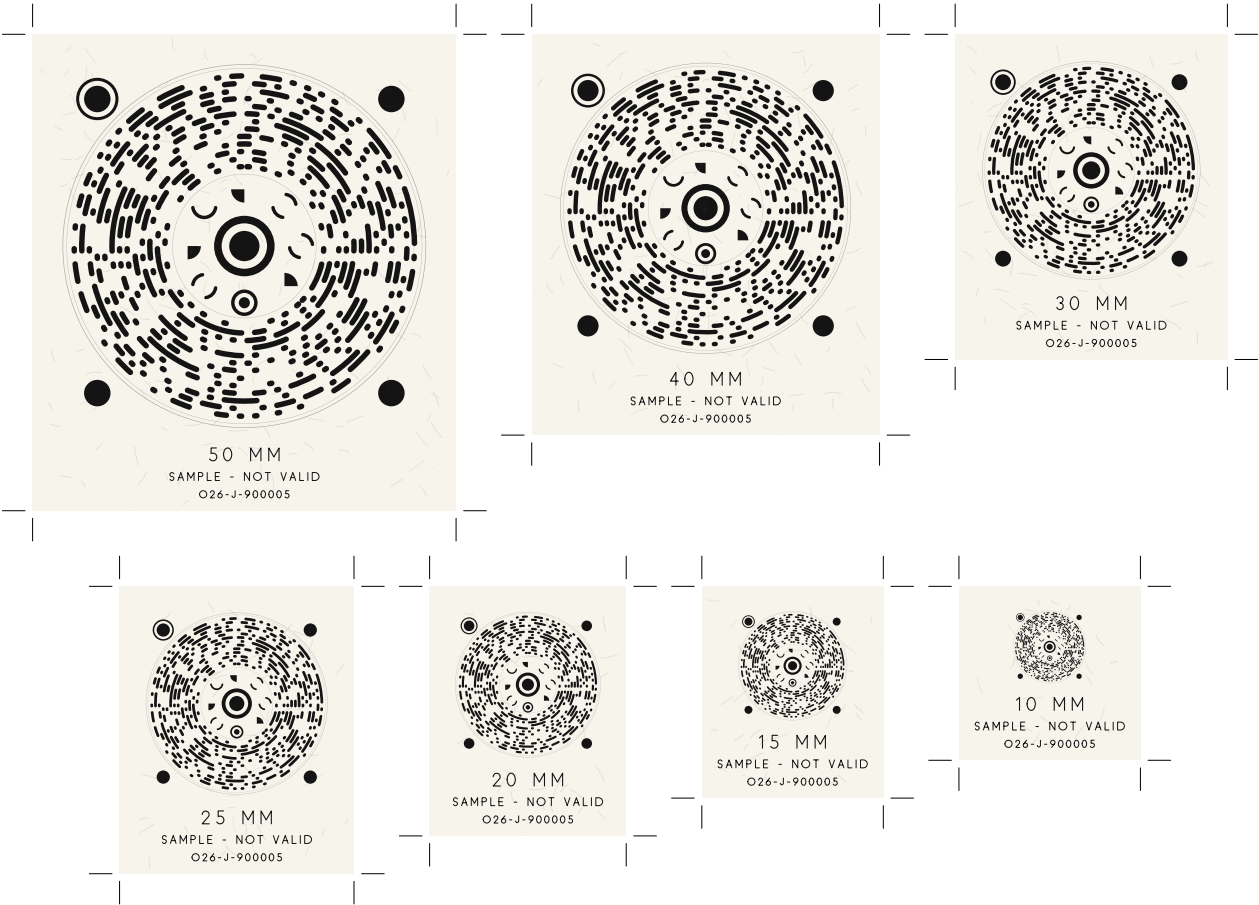
WRITE 3/3, 2/3, 1/3 OR 0/3 · LEAVE BLANK IF NOT TESTED · WRITE > WHEN THE CODE DOES NOT FIT INSIDE THE ON-SCREEN ORBIT AT THAT DISTANCE
DISTANCE = PHONE LENS TO TAG, PHONE PARALLEL TO THE TAG · A READ = ANY RESULT SCREEN WITHIN 10 S · CUT THE TAG OUT OR MASK THE OTHER CODES FIRST
THIS PAGE SIMULATES MATTE GREY OR RECYCLED STOCK ON PAPER · CONFIRM ON THE REAL MATERIAL BEFORE FIXING A MINIMUM SIZE

TEXTURED PAPER

RENDITION 5 / 7

FIBRE AND SPECK OVERLAY AT LOW CONTRAST, OVER THE CODE

SAMPLE CODE O26-J-900005 · KEY ID 1 · PUBLIC SAMPLE KEY · A VERIFIER MUST ANSWER INVALID SIGNATURE OR UNKNOWN



RESULTS · SUCCESSFUL READS OUT OF 3 ATTEMPTS PER CELL

PHONE / OS _____ BROWSER _____ DATE _____ LIGHT _____ PRINTER / DPI _____

DISTANCE	1X · NO ZOOM						2X ZOOM · IF THE SCANNER OFFERS IT					
	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM
10 MM												
15 MM												
20 MM												
25 MM												
30 MM												
40 MM												
50 MM												

WRITE 3/3, 2/3, 1/3 OR 0/3 · LEAVE BLANK IF NOT TESTED · WRITE > WHEN THE CODE DOES NOT FIT INSIDE THE ON-SCREEN ORBIT AT THAT DISTANCE

DISTANCE = PHONE LENS TO TAG, PHONE PARALLEL TO THE TAG · A READ = ANY RESULT SCREEN WITHIN 10 S · CUT THE TAG OUT OR MASK THE OTHER CODES FIRST

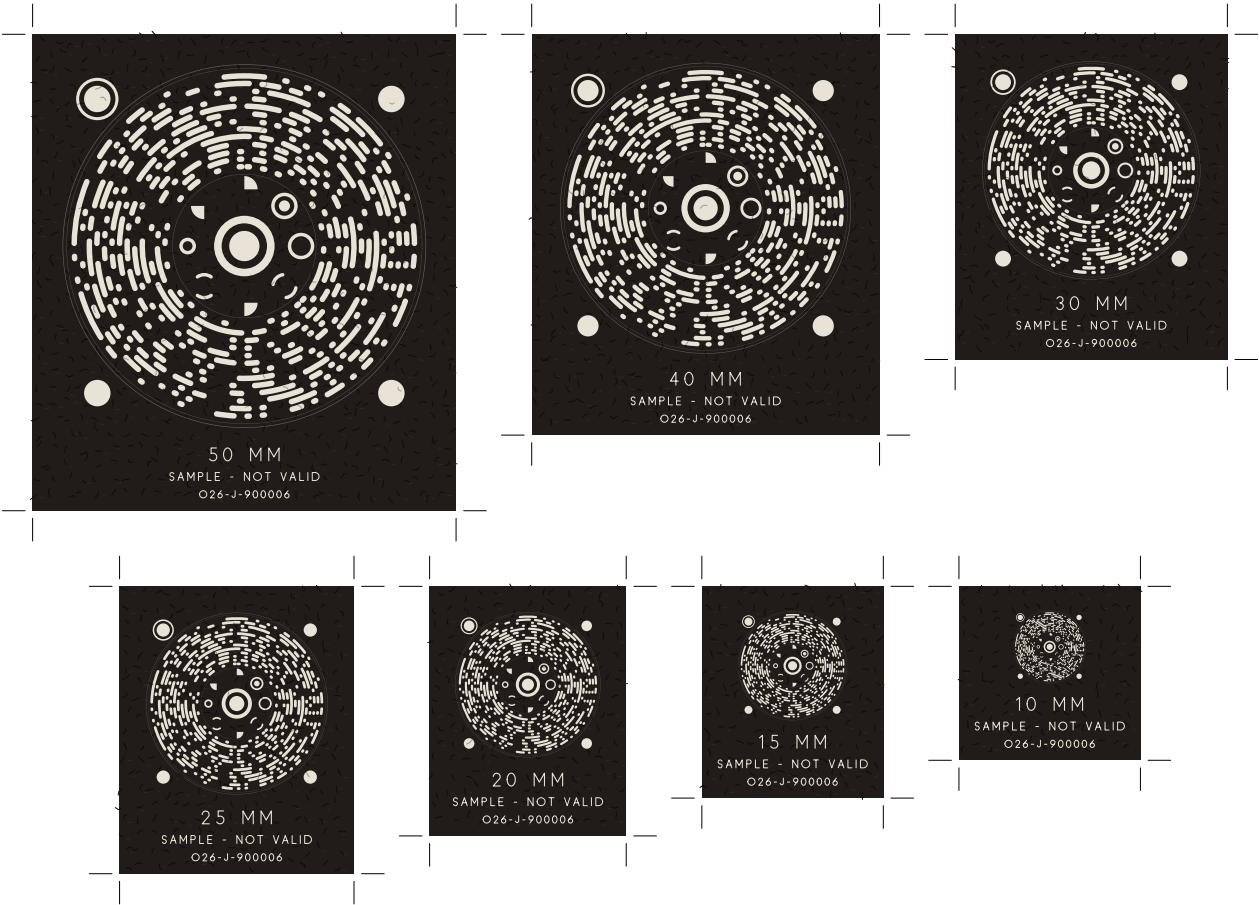
THIS PAGE SIMULATES COTTON, LAID OR FELT-MARKED PAPER ON PAPER · CONFIRM ON THE REAL MATERIAL BEFORE FIXING A MINIMUM SIZE

BLACK LEATHER

RENDITION 6 / 7

LIGHT INK #E9E3D7 ON GRAINED DARK SUBSTRATE #211C19

SAMPLE CODE O26-J-900006 · KEY ID 1 · PUBLIC SAMPLE KEY · A VERIFIER MUST ANSWER INVALID SIGNATURE OR UNKNOWN



RESULTS · SUCCESSFUL READS OUT OF 3 ATTEMPTS PER CELL

PHONE / OS _____ BROWSER _____ DATE _____ LIGHT _____ PRINTER / DPI _____

DISTANCE	1X · NO ZOOM						2X ZOOM · IF THE SCANNER OFFERS IT					
	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM
10 MM												
15 MM												
20 MM												
25 MM												
30 MM												
40 MM												
50 MM												

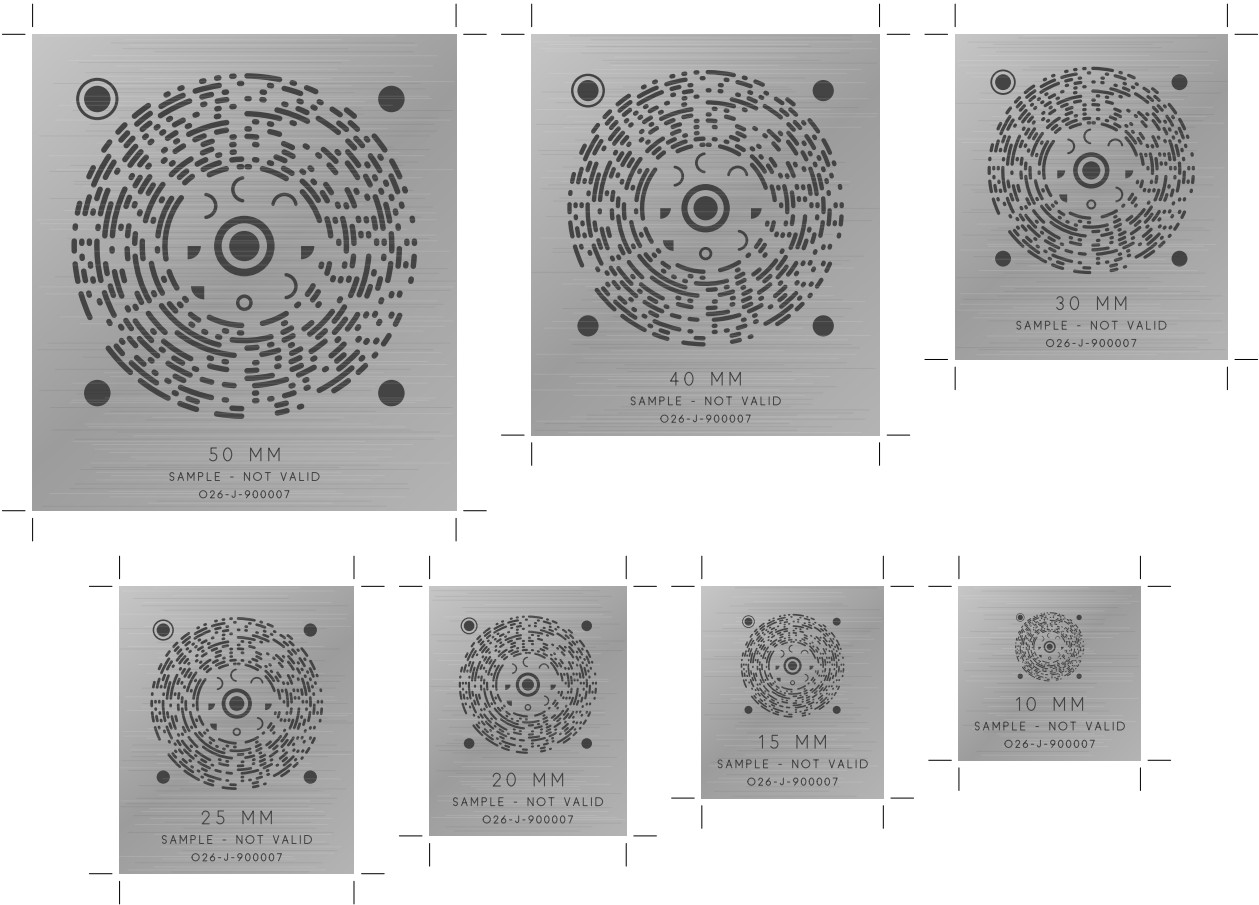
WRITE 3/3, 2/3, 1/3 OR 0/3 · LEAVE BLANK IF NOT TESTED · WRITE > WHEN THE CODE DOES NOT FIT INSIDE THE ON-SCREEN ORBIT AT THAT DISTANCE
DISTANCE = PHONE LENS TO TAG, PHONE PARALLEL TO THE TAG · A READ = ANY RESULT SCREEN WITHIN 10 S · CUT THE TAG OUT OR MASK THE OTHER CODES FIRST
THIS PAGE SIMULATES PRINTED OR FOILED DARK LEATHER ON PAPER · CONFIRM ON THE REAL MATERIAL BEFORE FIXING A MINIMUM SIZE

METALLIC SURFACE

RENDITION 7 / 7

DARKER ENGRAVING #444444 ON BRUSHED MID GREY · NO DECOR

SAMPLE CODE O26-J-900007 · KEY ID 1 · PUBLIC SAMPLE KEY · A VERIFIER MUST ANSWER INVALID SIGNATURE OR UNKNOWN



RESULTS · SUCCESSFUL READS OUT OF 3 ATTEMPTS PER CELL

PHONE / OS _____ BROWSER _____ DATE _____ LIGHT _____ PRINTER / DPI _____

DISTANCE	1X · NO ZOOM						2X ZOOM · IF THE SCANNER OFFERS IT					
	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM	8 CM	10 CM	12 CM	15 CM	20 CM	25 CM
10 MM												
15 MM												
20 MM												
25 MM												
30 MM												
40 MM												
50 MM												

WRITE 3/3, 2/3, 1/3 OR 0/3 · LEAVE BLANK IF NOT TESTED · WRITE > WHEN THE CODE DOES NOT FIT INSIDE THE ON-SCREEN ORBIT AT THAT DISTANCE
DISTANCE = PHONE LENS TO TAG, PHONE PARALLEL TO THE TAG · A READ = ANY RESULT SCREEN WITHIN 10 S · CUT THE TAG OUT OR MASK THE OTHER CODES FIRST
THIS PAGE SIMULATES LASER ENGRAVED STEEL, SILVER OR BRASS PLATES ON PAPER · CONFIRM ON THE REAL MATERIAL BEFORE FIXING A MINIMUM SIZE